Faint Pulse Source generation for fiber-based QKD systems

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Abstract

This project aimed to build and test a special light source needed for Quantum Key Distribution (QKD), a super-secure way to share secret keys. We used a specific type of 1550 nm DFB laser, common in internet fiber cables. We turned its steady light into very fast, ultra-dim flashes using a modulator and an attenuator. On average, each flash contained only about 0.2 photons, putting us in the quantum regime. We used precise timing equipment and custom Python code to check the system's performance. We confirmed that our electronic control pulses and the resulting light flashes were perfectly in sync and that the light's polarization was cleanly split.

A major finding was that our phase encoding—how we imprint the secret information onto the light—worked perfectly at a very slow speed (1 pulse per second) but failed at the fast speeds we need (1 million pulses per second and higher). The problem was a speed limit in our control electronics (the FPGA's digital-to-analog converter). This work is the first step in Pakistan toward building a practical QKD system and clearly shows that upgrading to a faster electronic converter is the key to making secure, high-speed quantum communication a reality.

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Chapter 1

Introduction

Quantum Key Distribution (QKD) represents a revolutionary cryptographic technique that enables two parties, traditionally designated as “Alice” (the sender) and “Bob” (the receiver), to generate and share a secret key with security guaranteed by the laws of quantum mechanics. The process involves Alice encoding random bits onto quantum states, typically the phase or polarization of single photons, and transmitting them to Bob. Bob then performs measurements to decode these states. The core security arises from fundamental quantum principles, namely, the no-cloning theorem and the observer effect, which ensure that any attempt by an eavesdropper to intercept or copy the key will inevitably introduce detectable disturbances into the transmission, alerting the legitimate users to the breach. As such, QKD is widely regarded as a vital solution to the future threat posed by quantum computers to current classical encryption algorithms like RSA.

The initial and critical step in any QKD system is the generation of the light source that carries the quantum information. While ideal systems employ true single-photon sources (e.g., quantum dots, NV centers), these are often complex, expensive, and difficult to integrate. Consequently, the vast majority of practical, fiber-based QKD systems utilize a pragmatic alternative: Weak Coherent Pulses (WCPs). These are pulses from a conventional laser that have been intensely attenuated to a level where the mean number of photons per pulse (μ) is significantly less than one. This approach effectively emulates a single-photon source while leveraging the robustness, maturity, and compatibility of standard telecommunication laser diodes and modulators.

In this project, our goal was to construct a fiber-based QKD source designed for eventual scalability to long-distance communication. We employed weak coherent pulses, targeting an attenuation to a mean photon number of approximately $\mu=0.25$, which firmly places the source into the desired quantum regime. The technical characteristics of these faint pulses, including their temporal width, wavelength stability, polarization state, and overall stability—are paramount determinants of both the system's performance and its ultimate security. Parameters like pulse duration directly influence timing resolution and error rates, while the chosen value of μ represents a critical trade-off between maintaining a viable key rate and minimizing vulnerabilities to specific quantum hacking strategies, such as photon-number-splitting (PNS) attacks.

To the best of my knowledge, this internship project constitutes the first dedicated experimental effort within Pakistan to generate and exhaustively characterize a faint pulse source explicitly for a fiber-based QKD system. This work is designed to address a significant research gap by laying the essential experimental groundwork required for the future development of practical quantum communication infrastructure within the country. The successful generation and detailed characterization of weak coherent pulses—encompassing accurate mean photon number

estimation, precise pulse shape analysis, and verification of polarization properties—provide an indispensable foundation for subsequent implementations of secure long-distance QKD in Pakistan.

Chapter 2

Theory and Experimental Components

The heart of our QKD transmitter is the photon source, which is a 14-pin Distributed Feedback (DFB) laser diode housed in a robust “butterfly” package. This laser operates in the telecom C-band at a wavelength of 1550 nm, a standard choice due to its low attenuation in optical fibers. Its working principle is based on stimulated emission of radiation. The integrated butterfly package is not merely a container; it is engineered to provide a highly stable operating environment. It incorporates a thermoelectric cooler (TEC) for active temperature stabilization, which is critical to prevent unwanted drifts in the laser's output wavelength and power. An integrated thermistor provides precise temperature monitoring, and a monitor photodiode allows for continuous feedback control of the laser diode's output power, enabling exceptionally precise and stable operation.

This DFB laser generates an intense and continuous beam of coherent photons. However, for QKD applications, a pulsed optical output is mandatory. To achieve this transformation, an intensity modulator (IM) is employed. The IM is a device fabricated from Lithium Niobate ($LiNbO_3$), a material with strong electro-optic properties, and is designed for 1550 nm operation. It functions based on the principle of Mach-Zehnder interference within a waveguide. The device modulates the intensity of the incoming continuous optical beam in direct response to electronic RF pulses applied to its electrodes. In our setup, we drove the modulator with electronic pulses of 4V amplitude, which were specifically chosen to be compatible with the 50-ohm impedance of the RF components and to ensure the modulator operates within its linear region for optimal fidelity and minimal distortion.

To prepare the now-pulsed light for subsequent polarization encoding, a 3-paddle mechanical polarization controller is utilized to meticulously adjust the state of polarization (SOP) of the light. This controller operates on the principle of stress-induced birefringence; as the fiber is coiled around the three paddles, mechanical stress alters the refractive index for different polarization axes. The first and third paddles function as Quarter Wave Plates (QWPs), each capable of rotating the polarization ellipse and converting between linear and circular polarization. The second, central paddle acts as a Half Wave Plate (HWP), which rotates the plane of linear polarization by approximately 90 degrees.

The optical fiber used throughout most of this setup is standard single-mode (SM) fiber, which is designed to propagate only one fundamental spatial mode. However, a key limitation of standard SM fiber is that it does not maintain the polarization state of photons due to random birefringence effects caused by imperfections and external stresses. Therefore, in sections where the preservation of polarization is absolutely critical, Polarization Maintaining (PM) fiber is used. PM fiber contains built-in stress rods that create a well-defined and dominant birefringence axis, allowing

only one principal polarization state to propagate efficiently while attenuating the other. This preserves the quantum information encoded in polarization. Due to its higher cost, PM fiber is deployed strategically rather than throughout the entire network.

The intensity modulator successfully creates optical pulses, but their intensity remains far too high for QKD, which requires weak coherent pulses with a mean photon number of less than one ($\mu < 1$) per pulse. To attenuate the light down to the genuine single-photon level, a variable optical attenuator is employed. This electronic component allows for precise, controlled reduction of the optical power, and thus the average number of photons per pulse, by over 100,000 times.

While this combination of components theoretically generates the desired weak optical pulses, their characterization requires specialized detectors. Classical photodiodes and digital oscilloscopes are entirely unsuitable for this task because their intensity threshold is far too high; they are designed to detect trillions of photons per pulse, whereas our signal consists of, at most, one photon per pulse. For initial alignment and visualization purposes at higher power, a homodyne detector can be used, as it can preserve the quantum information of the incoming light. However, for final operation and characterization at the single-photon level, an ID230 Infrared Single-Photon Detector is used. This is a free-running Indium Gallium Arsenide/Indium Phosphide (InGaAs/InP) avalanche photodiode (APD) photon counter that is thermoelectrically cooled to achieve an extremely low dark count rate. The detector operates in gated mode: it is biased above its breakdown voltage only for a brief window synchronized with the expected arrival time of a pulse. Upon the absorption of a single photon within this gate, it triggers a macroscopic, measurable electron avalanche. The detector counts and, crucially, the precise arrival times of these photons are acquired and visualized on a computer using a Time-Correlated Single Photon Counting (TCSPC) system. In our setup, we used a MultiHarp 150 TCSPC unit, which measures and histograms the incoming photons versus their arrival time with picosecond resolution.

For the phase-encoding stage of our research, a phase modulator (PM) is used to control the optical phase of the light pulses. This modulator encodes quantum information by imparting a specific phase shift (e.g., 0, $\pi/2$, π, $3\pi/2$) to individual optical pulses. This allows the sender, Alice, to transmit a secret key to the receiver, Bob, using quantum protocols like BB84 or Differential Phase Shift (DPS) QKD, which rely on phase differences to establish a shared key. The phase modulator must be carefully synchronized with the intensity modulator to ensure that the voltage-induced phase shift is applied at the exact moment the optical pulse from the IM arrives at the PM's waveguide.

Following the phase modulator, a Polarization Beam Splitter (PBS) is used to separate the incoming photons based on their polarization into two orthogonal components, directing each to a separate output port. The fundamental property of a PBS means that if one output port registers the maximum photon count for a specific polarization, the other port will simultaneously register the minimum count for the orthogonal polarization. By using a PBS in conjunction with two single-

photon detectors, we can observe the quantum correlations between these two orthogonal states and measure the system's error rate.

Furthermore, we also employed a standard 50:50 Intensity Beam Splitter (BS) with a pair of single-photon detectors to perform a Hanbury Brown and Twiss (HBT) experiment. This experiment is essential for studying the photon statistics of the source, particularly for measuring the second-order correlation function $g^2(0)$. A value of $g^2(0) < 1$ is a key signature for non-classical light and is a benchmark for distinguishing between attenuated laser light (which has g^(2)(0) = 1) and true single-photon sources.

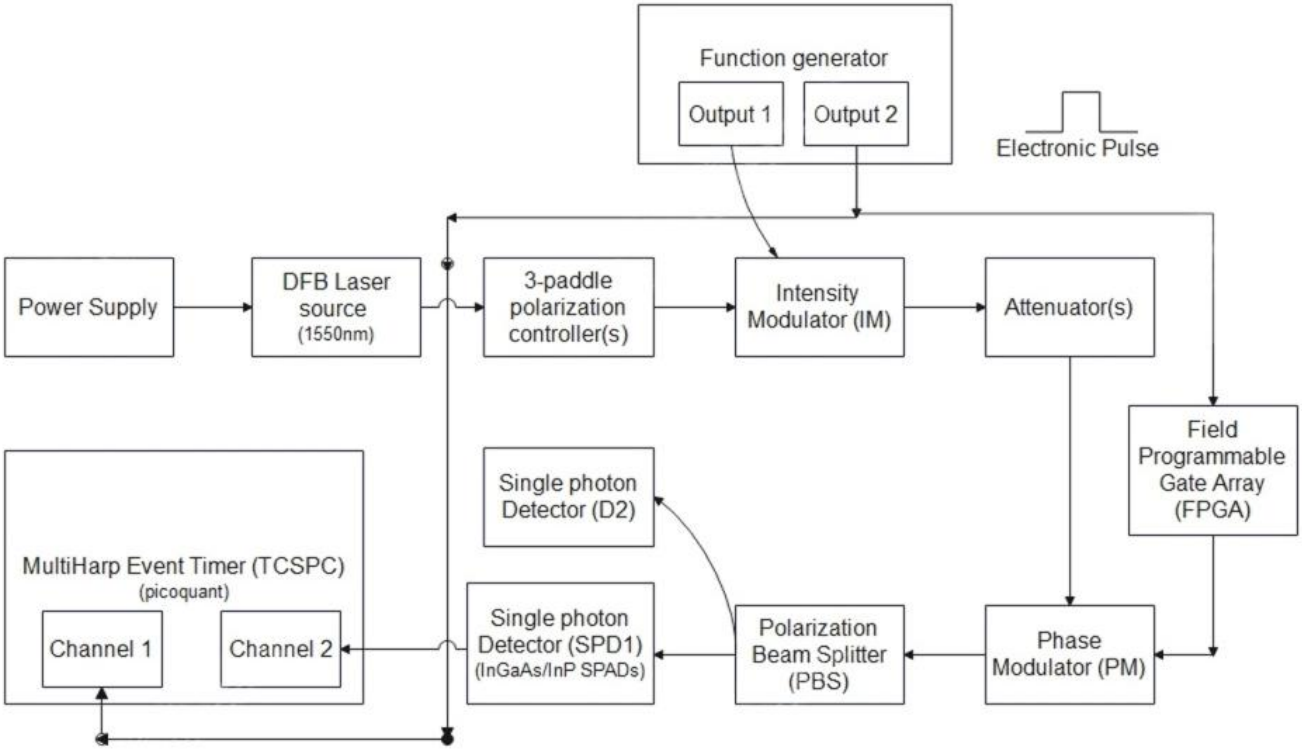


Fig 1. Schematic of the QKD source; (Components before FPGA are used for faint pulse-generation, Componets after FPGA are used for key-encoding.)

Chapter 3

Research Methodology

The MultiHarp TCSPC unit served as the primary data acquisition tool, providing a stream of data containing the precise picosecond-resolution arrival time of every detected photon. The analysis of this large and complex dataset was performed using custom software I wrote in Python. Python is exceptionally well-suited for this type of scientific data analysis due to its powerful, easy-to-use libraries such as NumPy for numerical computations, Pandas for data manipulation, Matplotlib for visualization, and SciPy for scientific algorithms. I developed specific scripts and logic to answer each of our core research questions, which are detailed below.

3.1 Characterizing the Intensity Modulator's Pulse Generation

Objective: To determine whether the intensity modulator was operating within its specified performance parameters for generating optical pulses, specifically evaluating metrics like the extinction ratio (the contrast between the 'on' and 'off' states), modulation depth, and rise/fall times.

Methodology: To analyze the pulsed output, I implemented a technique called modulo binning in Python. The program first read the absolute arrival time of each detected photon. Since the laser operates at a fixed repetition rate (e.g., 100 kHz or 1 MHz), its entire timeline can be divided into consecutive, identical periods. The modulo operation takes any photon's arrival time and calculates its remainder when divided by the laser's period. For example, for a 1 MHz repetition rate (period = 1 μs = 1,000,000 ps), a photon arriving at 1,000,127 ps and another at 2,000,127 ps would both be assigned to the same bin at 127 ps within the folded period. This effectively "folds" all photon arrivals from thousands of laser pulses into a single, representative period.

Implementation: A histogram was then created by counting how many photons arrived in each fine time bin within this folded period. The resulting plot provides a clear visual representation of the temporal shape of the optical pulse generated by the intensity modulator. A clean, narrow peak indicates the modulator is successfully creating distinct pulses with fast switching times, while a broad or poorly defined distribution suggests issues like slow switching, insufficient extinction (light leaking through in the 'off' state), or excessive noise.

Process: I systematically searched for the optimal operating parameters (RF pulse amplitude and DC bias voltage) at different frequencies (1 kHz, 100 kHz, 500 kHz, and 1 MHz) to produce the best possible pulse shape with the highest signal-to-noise ratio (SNR). This involved iterating

through various DC offset voltages to find the bias point that maximized the contrast between the pulse peak and the background.

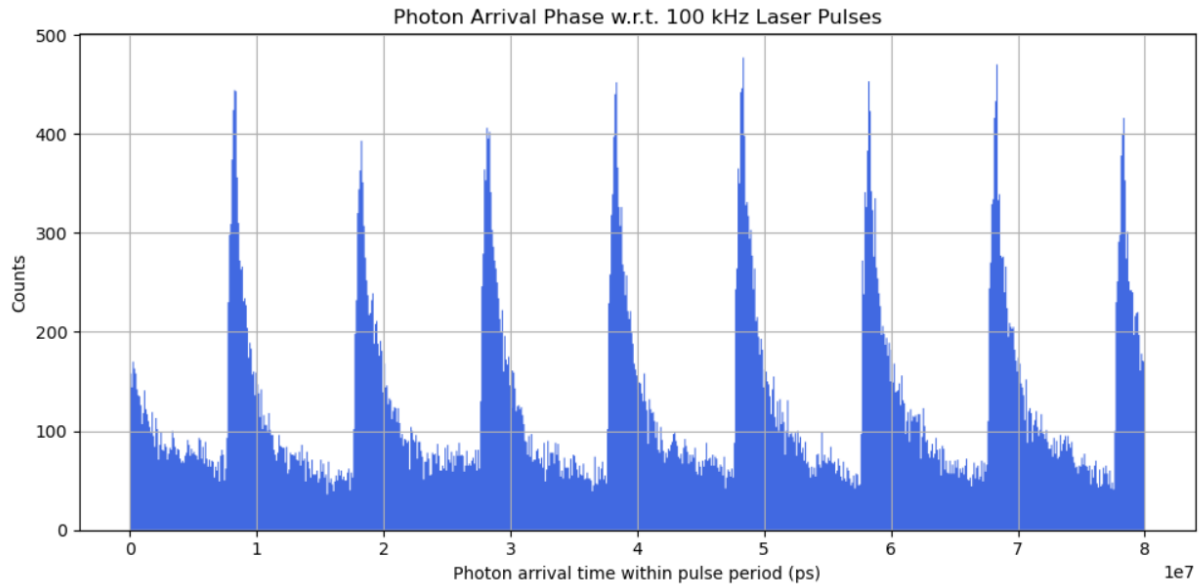


Fig 2. Modulo binning plot at frequency = 100kHz or pulse period= 1e7ps; pulse width =1µs (10% of the pulse period), V_{pp}= 4V with a DC offset of 915mV.

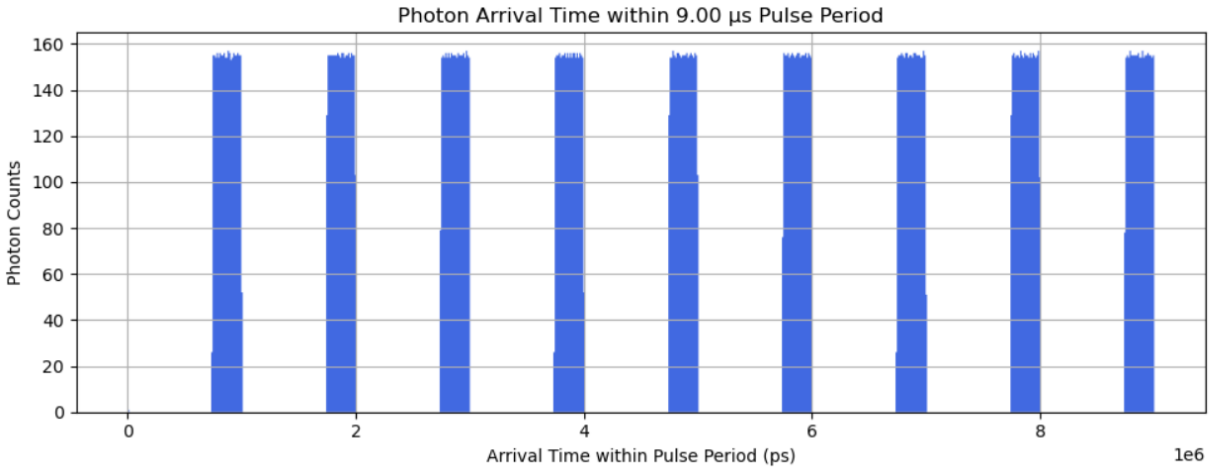


Fig 3. Modulo binning plot at frequency = 5MHz or pulse period= 1e6ps; pulse width =100ns (10% of the pulse period), V_{pp}= 4V with a DC offset of '-750mV'.

Note: The graphs provided in Fig 2. And Fig 3. are made using data files: "tel100" and "output_750.txt", this data was taken using a single output from function generator (towards intensity modulator) and the output of detector was simply directed towards a channel of event timer.

3.2 Quantifying Synchronization: Delay and Jitter Between Electronic and Optical Pulses

Objective: To precisely measure the temporal delay and the timing jitter (temporal uncertainty) between the electrical reference pulses that drive the system and the resulting optical pulses output from the intensity modulator. This synchronization is absolutely critical for the entire QKD protocol.

Methodology: I developed two complementary analytical methods in Python to investigate this:

1. **Macroscopic Temporal Coincidence Analysis:** The first program generated a combined histogram of event counts versus absolute time, plotting both the reference electronic pulses (on Channel 3 of the TCSPC) and the detected photon arrivals (on Channel 4) on the same set of axes. This visualization provided a direct, large-scale view of the system's operation over millions of pulses. The sustained presence of distinct photon peaks that align perfectly with the reference pulses over long measurement periods confirms successful and stable synchronization between the function generator and the IM.

2. **Precision Cross-Correlation Analysis:** The second program performed a more sophisticated and quantitative analysis. For every reference pulse timestamp, the algorithm searched for all photon detection events within a defined coincidence window (e.g., ±2000 ns) around it. It then constructed a histogram of the time differences (Δt = t_photon - t_reference) for all these events. This histogram was subsequently normalized by dividing by the average background count rate far from the peak. This yields a g^(2)-like correlation function. A sharp, prominent peak at $\Delta t = 0$ is the definitive signature of excellent synchronization. The narrow width of this peak (its Full Width at Half Maximum, or FWHM) directly reflects the combined timing jitter of the entire system, including the detectors and electronics. The value of the normalized function at the peak, g^(2)(0), provides insight into the photon statistics.

Challenges and Solutions: During this process, I identified and diagnosed several practical issues:

- **Impedance Mismatch:** Initially, using a single-output function generator with a passive T-connector to split the signal caused significant loading, degrading the pulse amplitude and shape at the event timer input. I resolved this first by inserting a high-impedance

amplifier and then definitively by upgrading to a multiple-output function generator, which provided isolated, pristine signals to each component.

- **Propagation Delays:** Differences in electrical cable lengths and internal processing times of components introduced nanosecond-scale delays, causing desynchronization. I characterized this macroscopically using a fast photodetector and an oscilloscope and then compensated by adding a programmable digital delay to the relevant function generator channel.

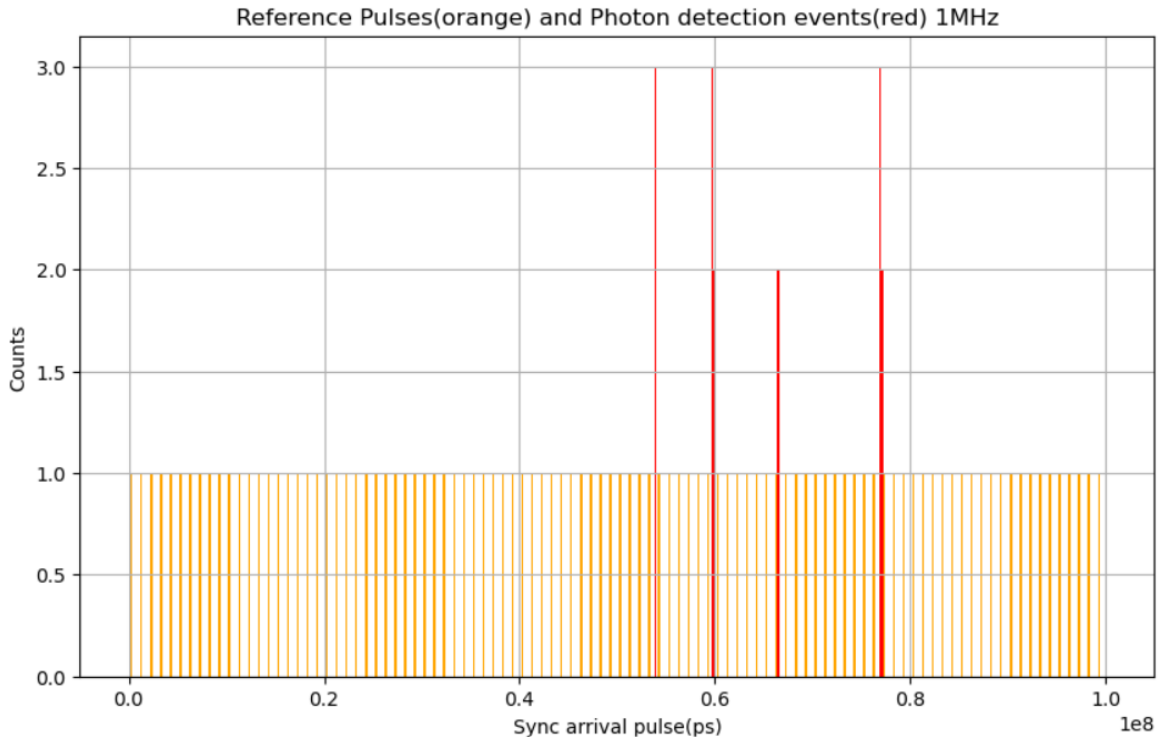


Fig 4. Counts per pulse plot showing the reference + photonic pulses; Pulse period = 1µs, pulse width = 100ns (10% of the pulse period), V_{pp} = 4V with a DC offset of -217mV.

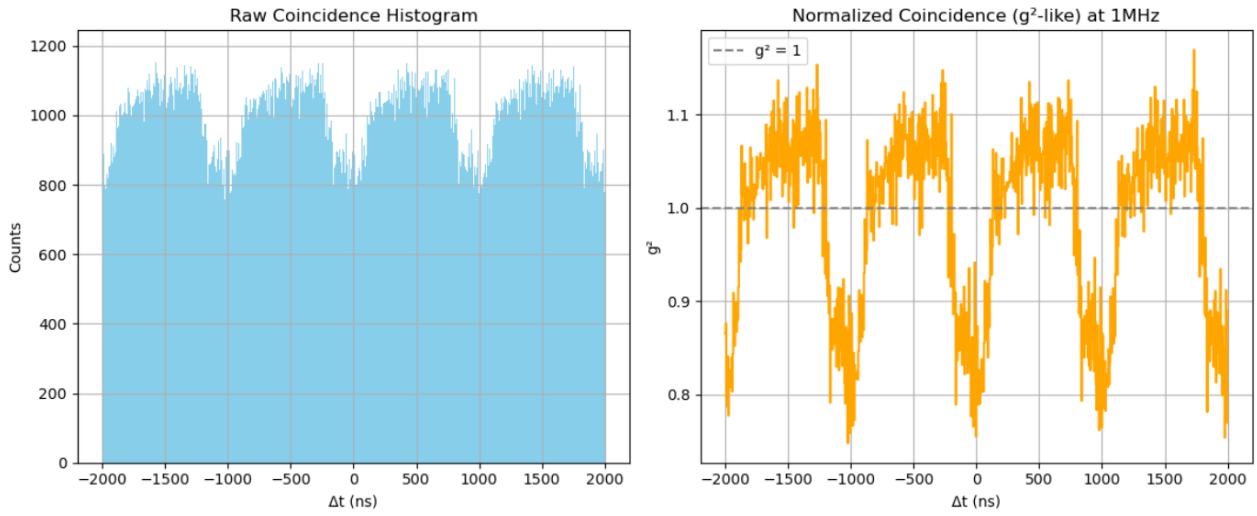


Fig 5. Correlations plot showing the raw and normalized coincidences; Pulse frequency= 1MHz, pulse width = 100ns (10% of the pulse period), V_{pp}= 4V with a DC offset of -217mV.

Note: The graphs/stats provided in Fig 4, Fig 5 and Fig 8 are made using data file: "output_217fullnew.txt", this data was taken using everything in the setup except FPGA where one output function generator was used with converter circuit and Tee. Two channels of event timer were used to compare detected and reference pulses.

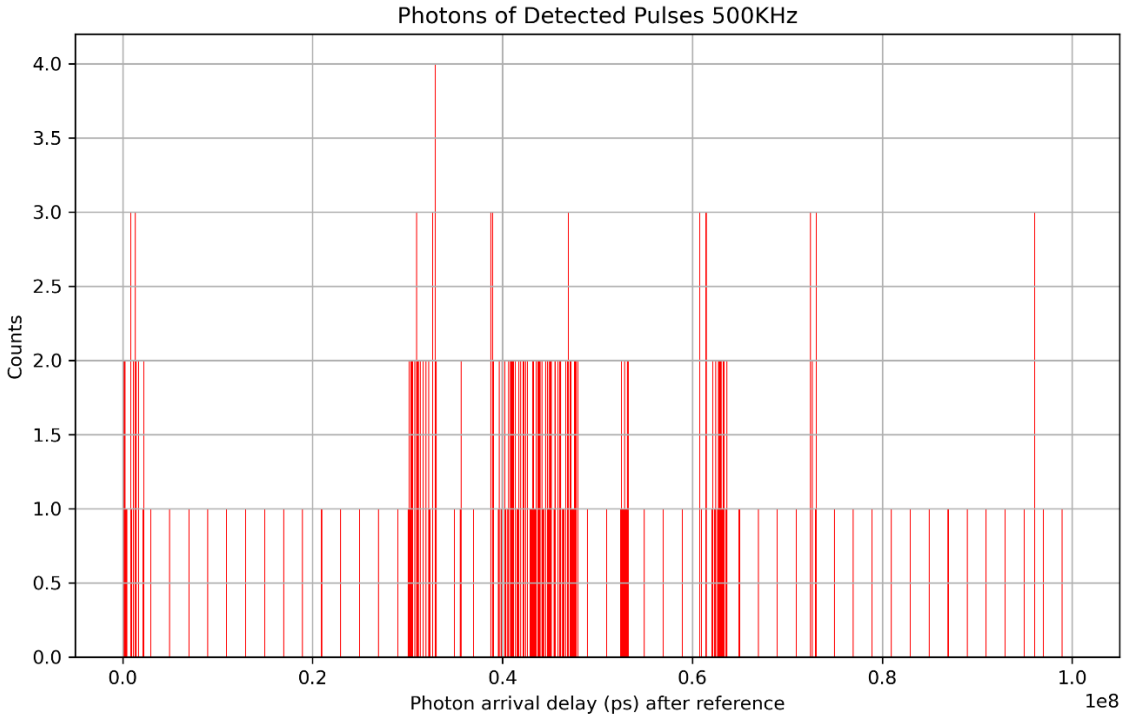


Fig 6. Counts per pulse plot showing the reference (1 count pulses) + photonic pulses; Pulse frequency = 500kHz, pulse width = 0.2μs (10% of the pulse period), V_{pp}= 4V with a DC offset of -217mV (File name= "output500kHz")

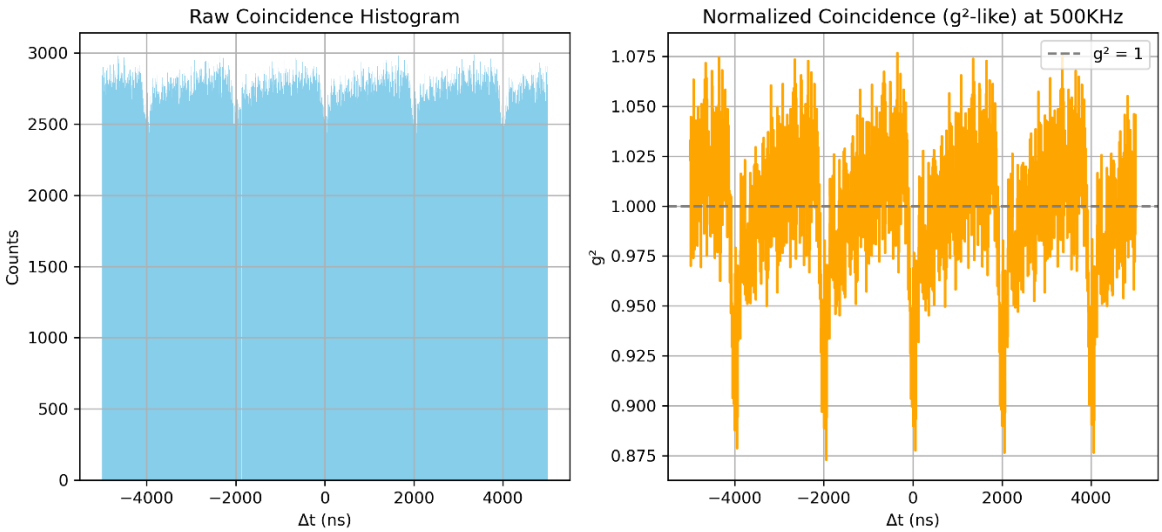


Fig 7. Correlations plot showing the raw and normalized coincidences; Pulse frequency= 500MHz, pulse width = 0.2μs (10% of the pulse period), V_{pp}= 4V with a DC offset of -217mV.

3.3 Calculating the Mean Photon Number and Attenuation

Objective: To accurately determine the mean photon number (μ) of the attenuated laser pulses, which is the average number of photons per pulse. This calculation must be corrected for the detector's dark counts to report the true signal value.
Methodology: The mean photon number per pulse was calculated using a procedure I coded in Python:

1. **Data Segmentation:** The photon arrival times (Channel 4) and the synchronous electronic trigger pulses (Channel 3) that mark the start of each laser cycle were separated from the raw time-tagged data.
2. **Pulse Window Definition:** The time period for each individual optical pulse was defined by the interval between two consecutive triggers on Channel 3. The first trigger signal in a pair (ref_times[i]) sets the start of the window, and the very next trigger (ref_times[i+1]) sets the end, effectively defining the bounds of the i-th pulse.
3. **Photon Counting per Window:** For each pulse window, the number of photons was counted. This was done efficiently by leveraging the fact that the photon timestamp array is sorted. For a given window [start, end], the algorithm finds the first photon index with a time >= start and the first photon index with a time >= end. The difference between these two indices yields the precise number of photons that arrived within that specific pulse window.
4. **Averaging and Dark Count Correction:** The raw mean photon number per pulse, μ_detected, was computed as the simple arithmetic average of the photon counts from all consecutive pulse windows. This value includes both signal photons and background dark counts. The average dark count rate per same-sized window was estimated from data taken with the laser off and subtracted to yield the final, corrected mean photon number, μ.
Theoretical Power Calculation: I also performed a theoretical calculation of the total attenuation. The input power (P_in) was measured directly with a power meter. The output power (P_out) was calculated from the measured photon rate and the energy per photon (hc/λ). The ratio P_out/P_in gave an attenuation factor, which was converted to decibels (dB), resulting in a value of approximately -98 dB, confirming the extreme attenuation required.

```
Max time (ps): 1000000000000.0
Total no of microsecond pulses counted: 948235
Mean photons per microsecond pulse: 0.211931
```

Fig 8. Mean photon no per microsecond pulse nearly equal to dark counts of the detector; Pulse frequency= 1MHz, pulse width = 100ns (10% of the pulse period), V_{pp} = 4V with a DC offset of -217mV.

Statistical Extraction of Signal from Noise

The raw stream of photon detection events is an amalgam of the desired signal—photons from the attenuated laser pulses—and inherent detector noise, primarily composed of dark counts and timing jitter. To isolate the true signal, we employed a statistical filtering technique predicated on the precise synchronization between the optical pulses and the electronic reference. The core logic is that genuine signal photons will arrive in a temporally correlated window immediately following each reference pulse, while noise events are randomly distributed across time.

Consequently, we defined a narrow coincidence window aligned with the expected arrival time of the optical pulse. Photon events falling within this window were classified as "signal," while those outsides were attributed to noise. This method confirmed the extreme attenuation necessary for a single-photon source: the vast majority of pulse windows were found to be empty, a signature of a true weak coherent pulse source where the probability of multi-photon events is negligible. The frequent empty windows directly validate that the mean photon number per pulse (μ) is significantly less than one, we achieved $\mu \approx 0.12$ (in the datafile “elc7” at 1MHz), which is a critical security requirement for the QKD protocol. This process allows us to quantitatively separate the faint quantum signal from the omnipresent classical noise of the detection system.

3.4 Measuring the Polarization Extinction Ratio of the Beam Splitter

Objective: To measure the Polarization Extinction Ratio (PER) between the two output ports of the PBS. The PER is a measure of how well a PBS separates polarization states, defined as the ratio of the power transmitted in the desired polarization to the power leaked in the orthogonal polarization. A high PER is crucial for minimizing errors in polarization-based QKD protocols. **Methodology:** I connected one output port of the PBS to one single-photon detector (Channel 2) and the other output port to a second detector (Channel 4). The TCSPC unit was then used to simultaneously record the counts from both detectors over time. **Result:** The real-time output and subsequent analysis showed perfect anti-correlation. When the polarization controller was adjusted so that one port (e.g., Channel 4) registered the maximum photon count for a specific input polarization, the other port (Channel 2) simultaneously registered only the detector's intrinsic dark count rate, which represents the minimum possible value. This stark contrast demonstrates a very high polarization extinction ratio, confirming that the PBS was performing as expected and that the polarization controller was able to effectively align the input state.

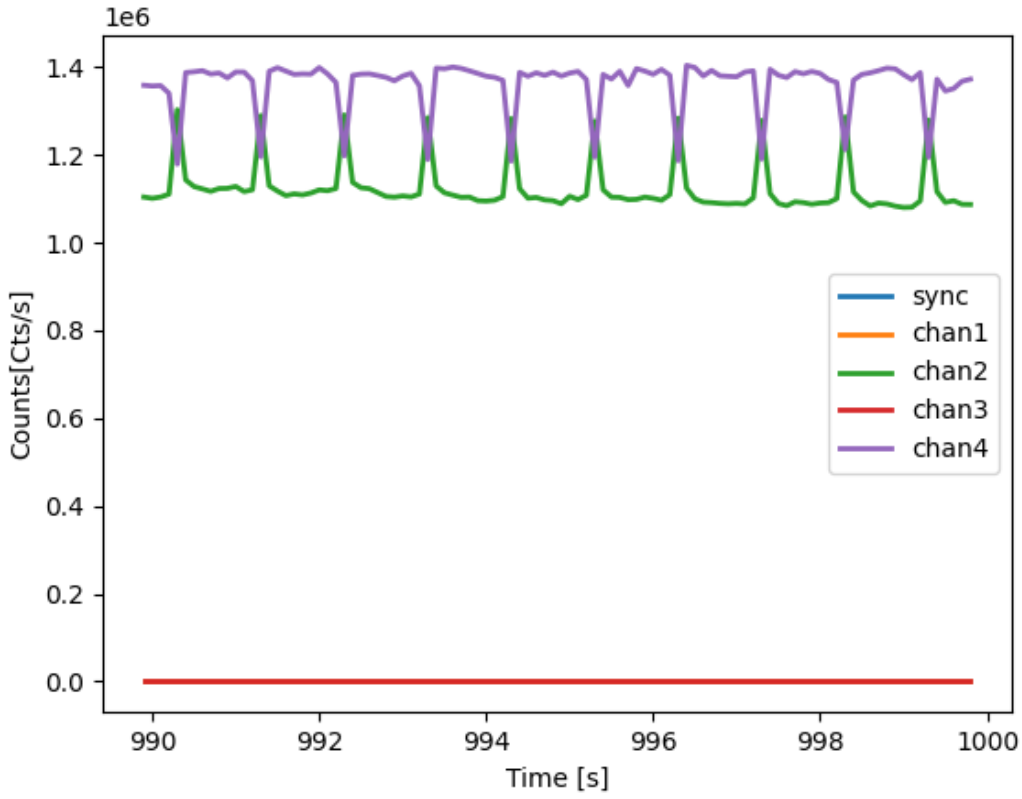


Fig 8. Real time output of MultiHarp's program (where the one port of PBS is connected to D1 showing chan 4 and the other port of PBS is connected to D2 showing chan 2); Pulse frequency = 1Hz

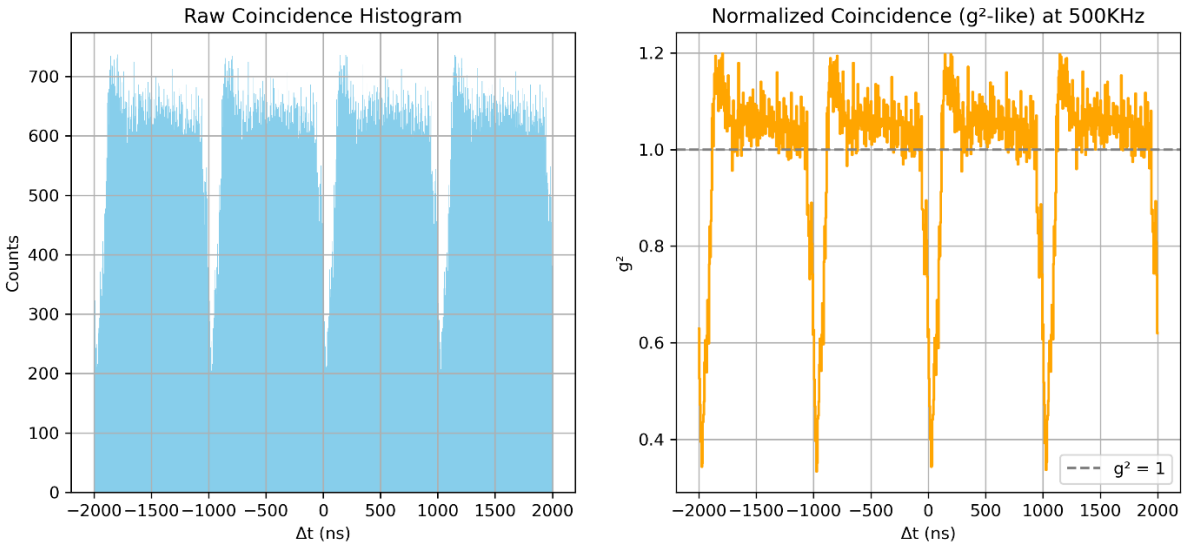


Fig 9. Correlation plot for port one of PBS.

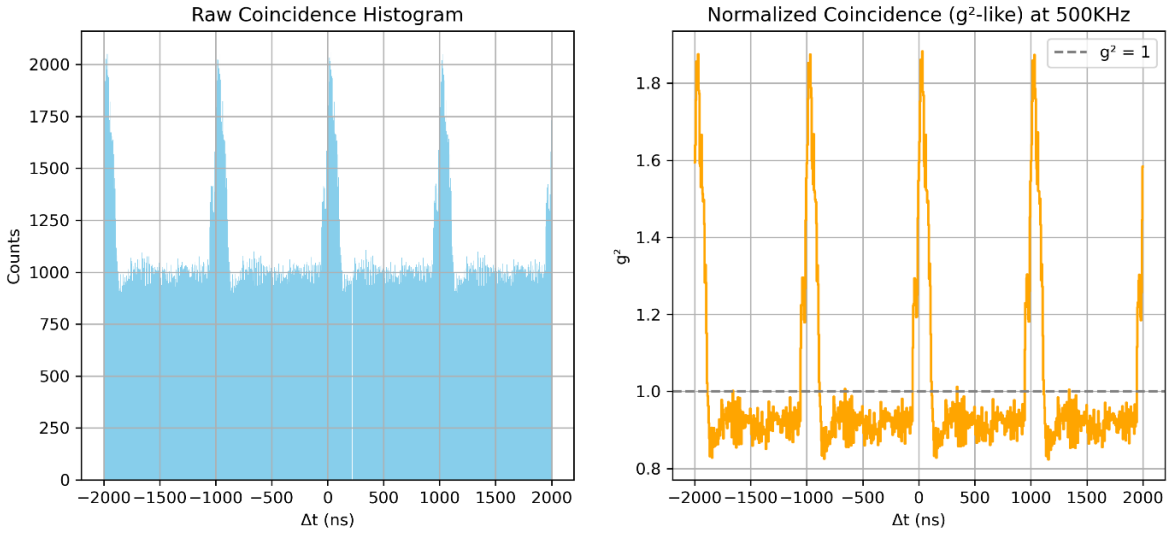


Fig 10. Correlation plot for port two of PBS.

3.5 Evaluating the Fidelity of the Phase Encoding System

Objective: This complex question had three parts: 1) To verify that the FPGA was generating the intended random voltage sequence with the correct amplitude and timing for the phase modulator. 2) To determine how faithfully the phase modulator translated these FPGA voltage levels into distinct and accurate optical phase shifts. 3) To measure the quantum bit error rate (QBER) resulting from this phase encoding process and assess if the distinction between the different quantum states was sufficient for secure key generation.
Methodology: To evaluate the entire encoding chain, I used a statistical approach based on the TCSPC data. The FPGA was programmed to output a predetermined random binary sequence (the "digital key"). The core logic of the BB84 protocol with phase encoding implies that a specific phase shift should deterministically route a pulse to a specific output port of the subsequent PBS. Therefore, for each time bin defined by a sync pulse, my analysis code assigned a binary value: a '1' was assigned if one or more photons were detected in the expected time window (indicating a phase shift leading to transmission), and a '0' was assigned if no photons were detected (indicating a phase shift leading to extinction). This process extracted a "photonic key" from the detector data. This photonic key was then directly compared, bit-by-bit, to the original "digital key" that was programmed into the FPGA. The ratio of mismatched bits to the total number of bits gives the Quantum Bit Error Rate (QBER).
Results and Analysis: The results were dramatically frequency-dependent:

- **At 1 Hz (Low Frequency):** The system demonstrated perfect fidelity. The photonic key extracted from the photon counts showed a 100% match with the FPGA's original key. A folded histogram of the photon arrivals showed two distinct levels: a near-zero count floor corresponding to '0' bits and a clear peak corresponding to '1' bits. This confirmed that the encoding logic, synchronization, and optical components were all functioning correctly.
- **At 1 MHz (High Frequency):** The system performance degraded completely. The extracted photonic key no longer correlated with the FPGA's key, resulting in a very high QBER. The folded histogram (see Fig. 12) lost its distinct two-level structure and instead showed a single, broadened peak, indicating that the phase modulator was no longer creating two distinguishable quantum states.
  Conclusion: The drastic difference in performance between low and high frequencies clearly identified the problem: the FPGA's digital-to-analog conversion circuitry lacked the necessary bandwidth to generate the precise, high-speed voltage transitions required by the phase modulator at MHz rates and beyond. The phase shifts were becoming smeared and inaccurate, rendering the encoding process unreliable.

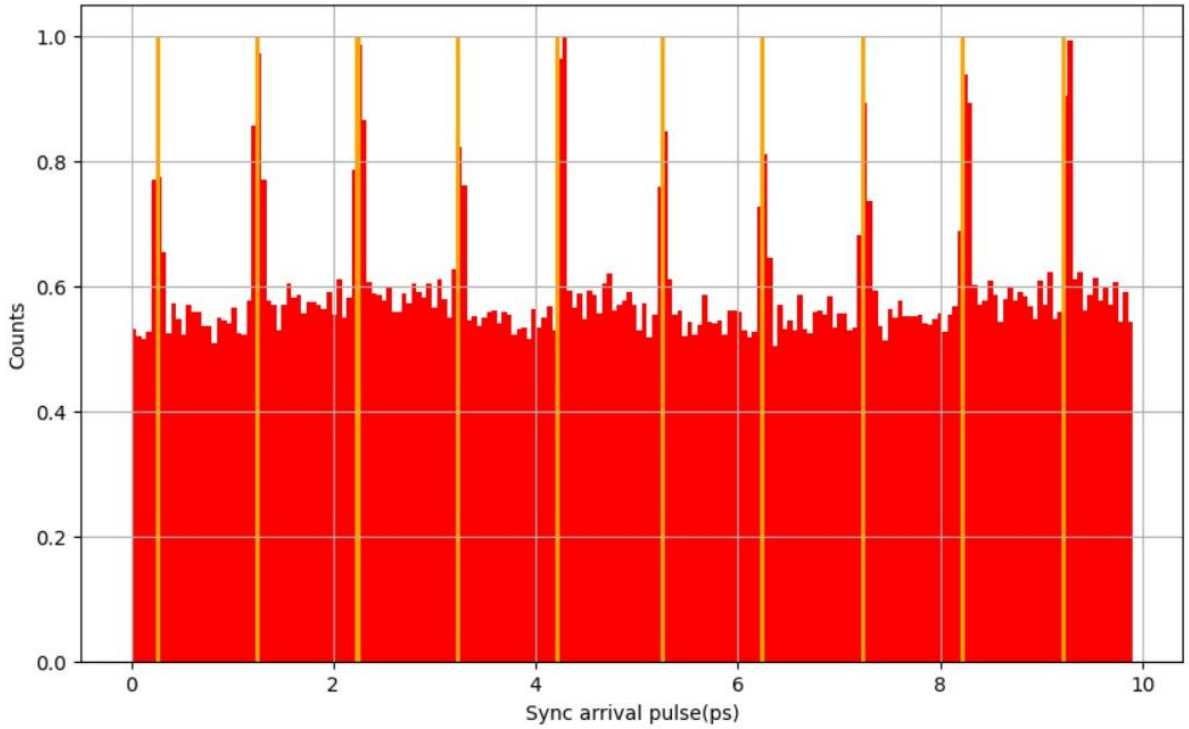


Fig 11. Normalized folded plot to compare the FPGA encoded peaks; Pulse frequency=1Hz, Max time= 10 seconds Data file used = "elc4"

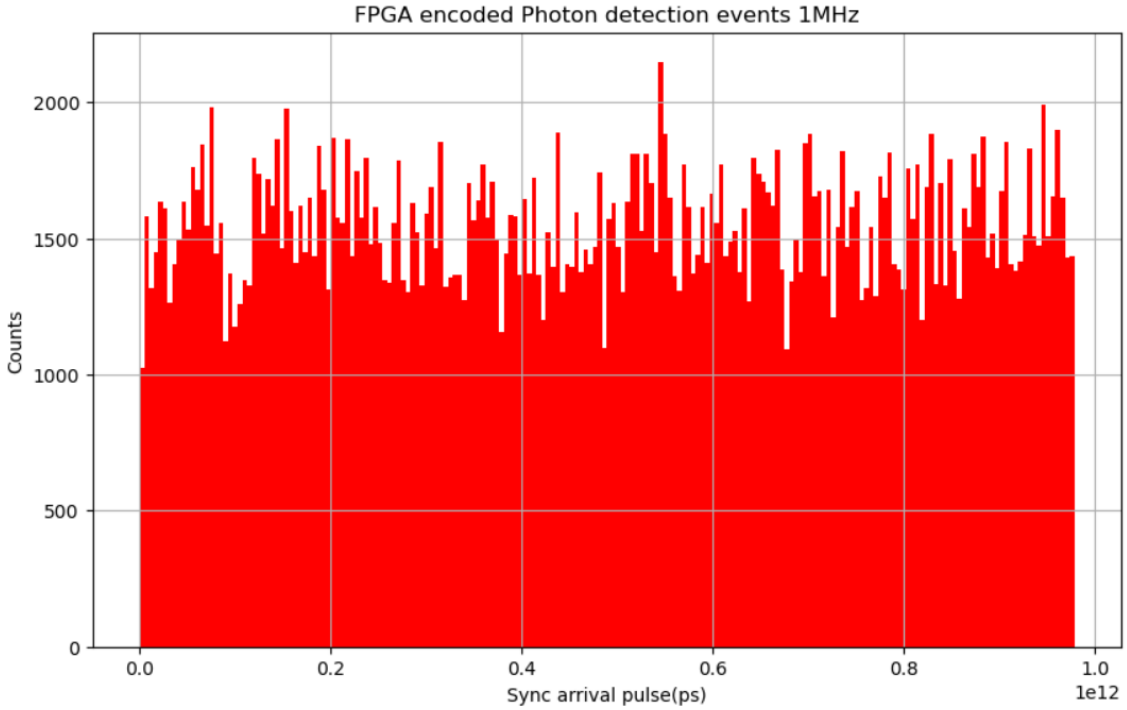


Fig 12. Un-normalized folded plot to compare the FPGA encoded peaks; Pulse frequency=1MHz, Max time= 1 second Data file used = "elc7"

Chapter 4

Discussion of results and outlook

4.1. Interpretation of Results and System Limitations

The results of this project demonstrate a successful proof-concept for the core faint pulse generation stage of a QKD system. The key achievement was the reliable production of weak coherent pulses at a telecommunications wavelength with a mean photon number of $\mu \approx 0.12$, which is an appropriate level for QKD protocols aiming to minimize multi-photon events. The comprehensive characterization confirmed several critical performance metrics: the intensity modulator was correctly biased to produce clean pulses, the synchronization between electronic and optical components was excellent with low jitter, and the polarization beam splitter was functioning with a high extinction ratio.

However, the investigation into the phase encoding system revealed a fundamental limitation that prevents high-speed operation. The perfect fidelity achieved at 1 Hz proves that the conceptual design, optical alignment, and core logic of the system are sound. The complete failure at 1 MHz is unambiguously traced to the digital-to-analog conversion stage of the FPGA. This component simply cannot switch between voltage levels quickly and precisely enough to define the distinct phase shifts required at these speeds. This bandwidth limitation is a common challenge when using general-purpose FPGAs for high-speed quantum control applications, as their primary design focus is digital logic, not analog signal integrity.

4.2. Future Outlook and Potential Improvements

The findings of this study provide a clear and actionable roadmap for transforming this proof-of-concept into a robust, high-speed, and secure QKD system. Future work should focus on the following strategic upgrades:

1. **High-Speed Digital-to-Analog Converter (DAC):** This is the most critical and immediate upgrade. Replacing the FPGA's analog output with a dedicated, high-resolution, high-speed DAC is essential. This specialized hardware is designed specifically for generating precise analog waveforms at GHz speeds, which would resolve the bandwidth bottleneck and enable faithful phase encoding at the target clock rates.

2. **Advanced Single-Photon Detection:** Employing superconducting nanowire single-photon detectors (SNSPDs) would offer significant advantages. SNSPDs typically have much lower dark count rates (<100 Hz), higher detection efficiencies (>90%), and significantly reduced timing jitter (<100 ps) compared to InGaAs APDs. This would dramatically improve the signal-to-noise ratio, allow for narrower coincidence windows, and enable more accurate QBER measurements.

3. **Enhanced System Stability:** Implementing active feedback systems is crucial for long-term operation. An automated polarization feedback system could continuously adjust the polarization controller to compensate for drift in the fiber. Active phase stabilization, often using a separate reference laser, could counteract phase fluctuations in the phase modulator arms, further reducing the error rate.
4. **Protocol-Level Security:** To enhance security against photon-number-splitting attacks, the system should be upgraded to implement the Decoy-State QKD protocol. This involves randomly alternating the intensity of the laser pulses (e.g., between signal, decoy, and vacuum levels), which allows Alice and Bob to detect eavesdropping attempts more effectively.
5. **Integrated Control and Modeling:** Developing a comprehensive software framework for real-time system control, simulation, and visualization would be invaluable. This would allow for monitoring of quantum states in real-time, rapid diagnostics of any issues, and optimization of system parameters for maximum performance and security

Conclusion

In conclusion, this project involved the successful assembly, integration, and detailed characterization of a faint pulse source for quantum key distribution. The work culminated in the generation of weak coherent pulses at 1550 nm with a mean photon number securely in the quantum regime. I developed and applied sophisticated data analysis techniques to verify the optimal performance of the individual components and their synchronization.

The most significant outcome was the clear identification of the system's current performance boundary: while the source itself is functional, the phase encoding fidelity is severely limited by the bandwidth of the control electronics at high speeds. This finding is not a failure but a critical result that provides a definitive direction for future research. By specifying the need for a high-speed DAC, this work lays a practical and foundational pathway for the continued development of quantum communication technology within Pakistan. It represents a crucial first step from theory towards a secure, functional quantum network.